State Estimation and Filtering for Robotic Perception

AA273

Factoring trajectories, SLAM, Factor graphs



Stanford University

MSL

AA273: State Estimation and Filtering for Robotic Perception

Announcements:

- HW 5 due Fri Feb 20
- HW 6 out Fri Feb 20, due Fri Mar 6
- Project Proposal: due Tue Feb 17
  - 2-3 pages, Latex
  - Related work >= 10 refs, ~1 page
  - Proposed work ~ 1 page
  - Bibtex refs

Lecture 14:

- Factoring trajectories
- SLAM
- Factor graphs

- Project Reports: due Tue Mar 17
  - 5-6 pages + 1 page per group member > 1
  - Latex/Overleaf, IEEE conf format
  - >= 10 refs
  - Describe research and results
  - Re-use proposal material
- Project Presentations: 3/10, 3/12
  - 5 min pres + 3 min Q and A
  - Sign up for preferred time slot
  - Share slides through zoom meeting
  - Present on your computer

Midterm grades

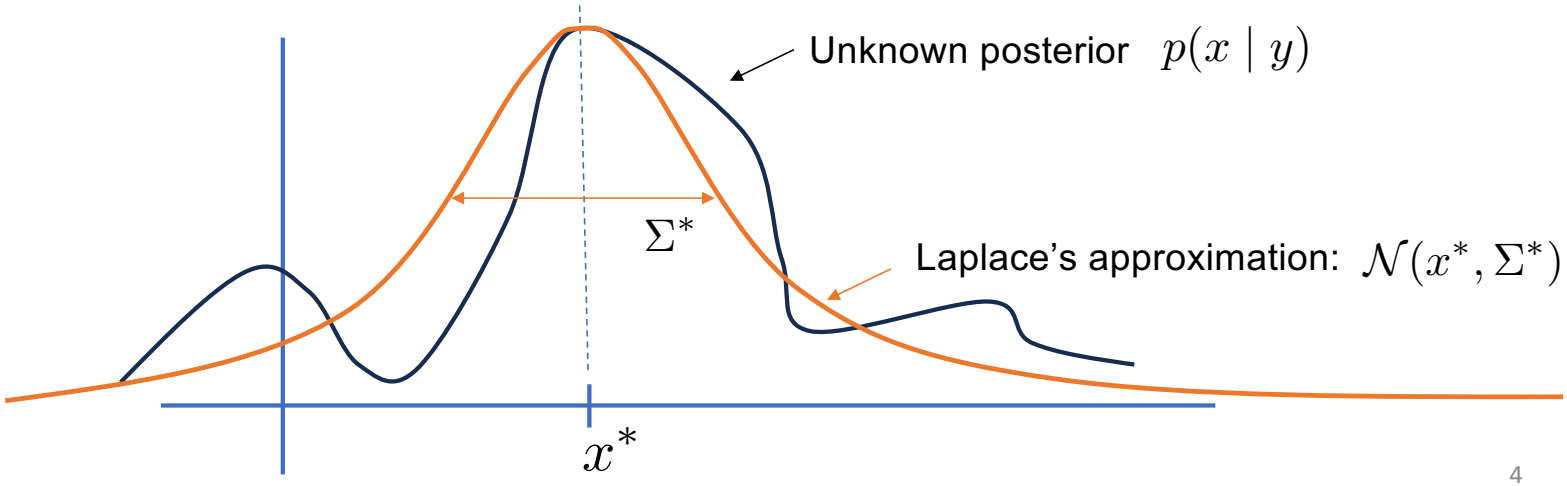
Statistics:
Minimum: 48.0
Maximum: 105.0
Mean: 91.47
Median: 97.0
Standard Deviation: 14.4

Recall: Solving for Laplace's Approximation

Suppose we can't find posterior directly: $p(x \mid y)$

Approximate with Gaussian Laplace Approx.:

$$p(x \mid y) \approx \mathcal{N}(x^*, \Sigma^*) \begin{cases} x^* = \arg\min_x -\log(p(x \mid y)) \\ \Sigma^* = \left[\nabla_x^2 - \log(p(x \mid y))\right]^{-1} \end{cases}$$



Recall: Nonlinear least squares formulation

Bayes' Rule: $p(x \mid y) = \dfrac{p(y \mid x)p(x)}{p(y)}$

With nonlinear-Gaussian likelihood model and Gaussian prior, we have:

Likelihood:

$$Y = g(X) + V \qquad V \sim \mathcal{N}(0, R)$$

$$-\log(p(y \mid x)) \propto \frac{1}{2}\|y - g(x)\|^2_{R^{-1}}$$

Prior:

$$X \sim \mathcal{N}(\mu_x, \Sigma_x)$$

$$-\log(p(x)) \propto \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}}$$

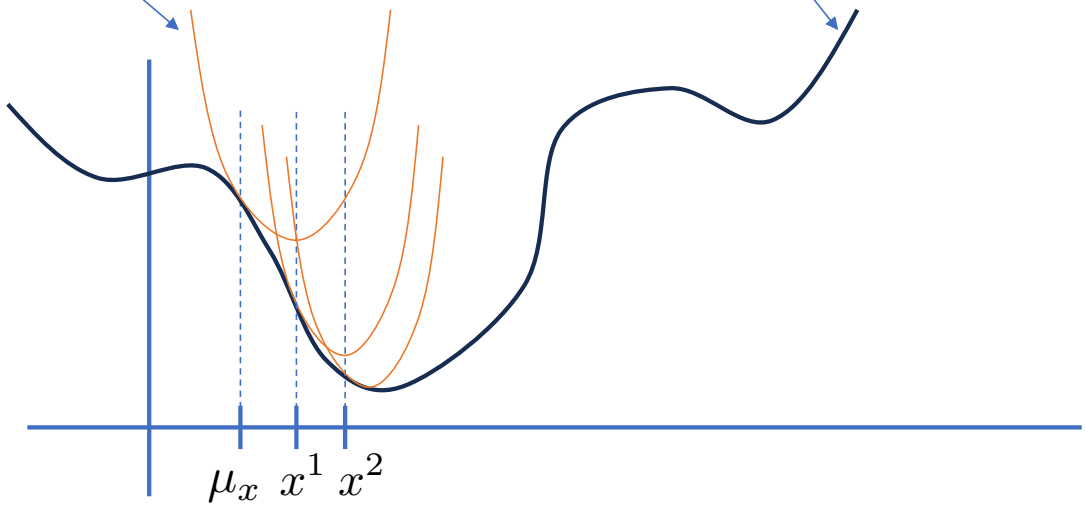
Nonlinear least squares:

$$x^* = \arg\min_x \frac{1}{2}\|y - g(x)\|^2_{R^{-1}} + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}}$$

Recall: Gauss-Newton optimization

$$x^* = \arg\min_x \frac{1}{2}\|y - g(x)\|^2_{R^{-1}} + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}}$$

$$x^k = \arg\min_x \frac{1}{2}\|y - (g(x^{k-1}) + C^{k-1}(x - x^{k-1}))\|^2_{R^{-1}} + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}}$$



Iterative EKF (iEKF) Algorithm

$$x^* = \arg\min_x \frac{1}{2}\|y - g(x)\|^2_{R^{-1}} + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}}$$

Initialize with: $x^0 = \mu_x$

Iterate: k = 1, 2, 3, …

$$C^{k-1} = \nabla_x g(x^{k-1})$$

$$x^k = \arg\min_x \frac{1}{2}\|\overbrace{(y - g(x^{k-1}) + C^{k-1}x^{k-1})}^{y^{k-1}} - C^{k-1}x\|^2_{R^{-1}} + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}}$$

$$x^k = \arg\min_x \frac{1}{2}\|y^{k-1} - C^{k-1}x\|^2_{R^{-1}} + \frac{1}{2}\|x - \mu_x\|^2_{\Sigma_x^{-1}}$$ Linear least squares

Solution:

$$x^k = \mu_x + K^{k-1}(y - g(x^{k-1}) - C^{k-1}(\mu_x - x^{k-1}))$$
$$\Sigma^k = \Sigma_x - K^{k-1}C^{k-1}\Sigma_x$$
$$K^{k-1} = \Sigma_x {C^{k-1}}^T (C^{k-1}\Sigma_x {C^{k-1}}^T + R)^{-1}$$

Iterative EKF (iEKF) updates

Repeat to convergence

Factoring the Full Trajectory

Consider the full distribution over states and measurements:

$$
\begin{aligned}
(X_{0:T} \mid Y_{1:T}) \sim p(x_{0:T} \mid y_{1:T}) &\propto p(x_{0:T}, y_{1:T}) \\
&= p(y_T \mid x_{0:T}, y_{1:T-1}) p(x_{0:T}, y_{1:T-1}) \\
&\qquad \text{Cond. Indep. Measurements} \\
&= p(y_T \mid x_T) \left(p(x_T \mid x_{0:T-1}, y_{1:T-1}) p(x_{0:T-1}, y_{1:T-1}) \right) \\
&\qquad \text{Markovianity} \\
&= p(y_T \mid x_T) p(x_T \mid x_{T-1}) p(x_{0:T-1}, y_{1:T-1}) \\
&\vdots \qquad \text{Recursion} \\
&= \left(\prod_{t=1}^{T} p(y_t \mid x_t) p(x_t \mid x_{t-1}) \right) p(x_0)
\end{aligned}
$$

Laplace's Approximation over Full Trajectory

$$p(x_{0:T} \mid y_{1:T}) \approx \mathcal{N}(x^*_{0:T}, \Sigma^*_{0:T})$$ Where:

$$x^*_{0:T} = \arg\min_{x_{0:T}} \left(\sum_{t=1}^{T} -\log(p(y_t \mid x_t)) - \log(p(x_t \mid x_{t-1})) \right) - \log(p(x_0))$$

$$\Sigma^*_{0:T} = \nabla^2_{x_{0:T}} \left[\arg\min_{x^*_{0:T}} \left(\sum_{t=1}^{T} -\log(p(y_t \mid x^*_t)) - \log(p(x^*_t \mid x^*_{t-1})) \right) - \log(p(x^*_0)) \right]$$

Consider:

Nonlinear-Gaussian Measurement Model:

$$Y_t = g(X_t) + V_t \qquad V_t \sim \mathcal{N}(0, R_t)$$

$$-\log(p(y_t \mid x_t)) = \frac{1}{2}\|y_t - g(x_t)\|^2_{R_t^{-1}}$$

Nonlinear-Gaussian Dynamics Model:

$$X_t = f(X_{t-1}, u_{t-1}) + W_{t-1} \qquad W_{t-1} \sim \mathcal{N}(0, Q_t)$$

$$-\log(p(x_t \mid x_{t-1})) = \frac{1}{2}\|x_t - f(x_{t-1}, u_{t-1})\|^2_{Q_t^{-1}}$$

Nonlinear Gaussian over Full Trajectory

$$p(x_{0:T} \mid y_{1:T}) \approx \mathcal{N}(x^*_{0:T}, \Sigma^*_{0:T})$$

Where:

$$x^*_{0:T} = \arg\min_{x_{0:T}} \sum_{t=1}^{T} \left(\frac{1}{2} \|y_t - g(x_t)\|^2_{R_t^{-1}} + \frac{1}{2} \|x_t - f(x_{t-1}, u_{t-1})\|^2_{Q_t^{-1}} \right) + \frac{1}{2} \|x_0 - \mu_0\|^2_{\Sigma_0^{-1}}$$

$$\Sigma^*_{0:T} = \nabla^2_{x_{0:T}} \left[\sum_{t=1}^{T} \left(\frac{1}{2} \|y_t - g(x_t)\|^2_{R_t^{-1}} + \frac{1}{2} \|x_t - f(x_{t-1}, u_{t-1})\|^2_{Q_t^{-1}} \right) + \frac{1}{2} \|x_0 - \mu_0\|^2_{\Sigma_0^{-1}} \right]$$

Can solve with Gauss-Newton or OTS nonlinear optimization solver (e.g., Gurobi)

$x^*_{0:T}$ Called **smoothed** trajectory estimate

x^*_t Similar to EKF mean, but with **smaller covariance** Σ^*_t

Uses **measurements in the future** $y_\tau, \tau > t$, to improve state estimate at x_t

Factor Graph Interpretation

$$-\log(p(x_{0:T} \mid y_{1:T})) = \sum_{t=1}^{T} \left(\frac{1}{2}\|y_t - g(x_t)\|^2_{R_t^{-1}} + \frac{1}{2}\|x_t - f(x_{t-1}, u_{t-1})\|^2_{Q_t^{-1}} \right) + \frac{1}{2}\|x_0 - \mu_0\|^2_{\Sigma_0^{-1}}$$

Measurement factor

Dynamics factor

Prior factor

See details on the board and in the notes.

SLAM Problem Formulation

See details on the board and in the notes.